5.2 Limits and Derivatives

In this section, we will discuss about limits, continuity of a function, differentiability and differentiation from first principle. Limits are used when we have to find the value of a function near to some value.

Definition 5.1

Let f be a real valued function of x. Let l and a be two fixed numbers. If $f(x)$ approaches the value l as x approaches to a, then we say that l is the limit of the function $f(x)$ as x tends to a, and written as

$$\lim_{x \to a} f(x) = l$$

Notations:

(i) $L[f(x)]_{x=a} = \lim_{x \to a^-} f(x)$ is known as left hand limit of $f(x)$ at $x=a$.

(ii) $R[f(x)]_{x=a} = \lim_{x \to a^+} f(x)$ is known as right hand limit of $f(x)$ at $x=a$.

5.2.1 Existence of limit

$\lim_{x \to a} f(x)$ exist if and only if both $L[f(x)]_{x=a}$ *and* $R[f(x)]_{x=a}$ exist and are equal.

i.e., $\lim_{x \to a} f(x)$ exist if and only if $L[f(x)]_{x=a} = R[f(x)]_{x=a}$.

NOTE

$L[f(x)]_{x=a}$ and $R[f(x)]_{x=a}$ are shortly written as $L[f(a)]$ and $R[f(a)]$

5.2.2 Algorithm of left hand limit: $L[f(x)]_{x=a}$

(i) Write $\lim_{x \to a^-} f(x)$.

(ii) Put $x=a-h$, $h>0$ and replace $x \to a^-$ by $h \to 0$.

(iii) Obtain $\lim_{h \to 0} f(a-h)$.

(iv) The value obtained in step (iii) is called as the value of left hand limit of the function $f(x)$ at $x=a$.

5.2.3 Algorithm of right hand limit: $R[f(x)]_{x=a}$

(i) Write $\lim_{x \to a^+} f(x)$.

(ii) Put $x = a+h$, $h>0$ and replace $x \to a^+$ by $h \to 0$.

(iii) Obtain $\lim_{h \to 0} f(a+h)$.

(iv) The value obtained in step (iii) is called as the value of right hand limit of the function $f(x)$ at $x=a$.

Example 5.14

Evaluate the left hand and right hand limits of the function

$$f(x) = \begin{cases} \dfrac{|x-3|}{x-3} & \text{if } x \neq 3 \\ 0 & \text{if } x = 3 \end{cases} \quad \text{at } x = 3$$

Solution

$$\begin{aligned} L[f(x)]_{x=3} &= \lim_{x \to 3^-} f(x) \\ &= \lim_{h \to 0} f(3-h),\ x = 3-h \\ &= \lim_{h \to 0} \frac{|(3-h)-3|}{(3-h)-3} \\ &= \lim_{h \to 0} \frac{|-h|}{-h} \\ &= \lim_{h \to 0} \frac{h}{-h} \end{aligned}$$

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